

Problem

Consider the full-wave rectified sinusoidal current in Table 17.3. Assume that the current is passed through a $1\text{-}\Omega$ resistor.

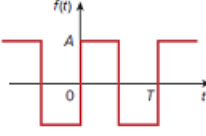
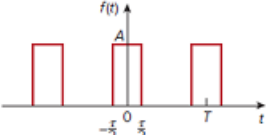
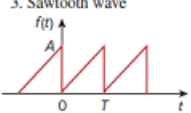
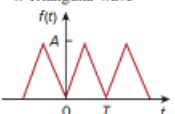
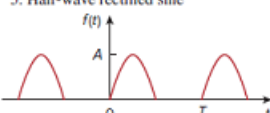
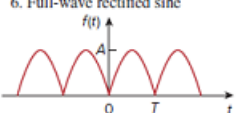
(a) Find the average power absorbed by the resistor.

(b) Obtain cn for $n = 1, 2, 3$, and 4 .

(c) What fraction of the total power is carried by the dc component?

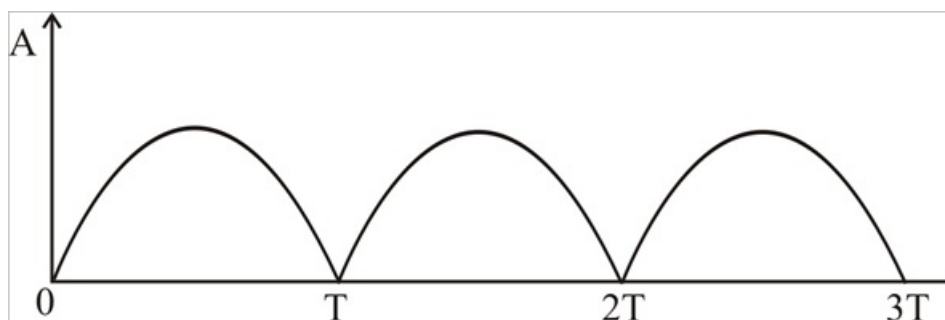
(d) What fraction of the total power is carried by the second harmonic ($n = 2$)?

Table 17.3: The Fourier series of common functions.

Function	Fourier series
1. Square wave 	$f(t) = \frac{4A}{\pi} \sum_{n=1}^{\infty} \frac{1}{2n-1} \sin(2n-1)\omega_0 t$
2. Rectangular pulse train 	$f(t) = \frac{A\tau}{T} + \frac{2A}{T} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi\tau}{T} \cos n\omega_0 t$
3. Sawtooth wave 	$f(t) = \frac{A}{2} - \frac{A}{\pi} \sum_{n=1}^{\infty} \frac{\sin n\omega_0 t}{n}$
4. Triangular wave 	$f(t) = \frac{A}{2} - \frac{4A}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \cos(2n-1)\omega_0 t$
5. Half-wave rectified sine 	$f(t) = \frac{A}{\pi} + \frac{A}{2} \sin \omega_0 t - \frac{2A}{\pi} \sum_{n=1}^{\infty} \frac{1}{4n^2-1} \cos 2n\omega_0 t$
6. Full-wave rectified sine 	$f(t) = \frac{2A}{\pi} - \frac{4A}{\pi} \sum_{n=1}^{\infty} \frac{1}{4n^2-1} \cos n\omega_0 t$

Step-by-step solution

Step 1 of 5



Step 2 of 5

$$f(t) = \frac{2A}{\pi} - \frac{4A}{\pi} \sum_{n=1}^{\infty} \frac{1}{4n^2 - 1} \cos(n\omega_o t)$$

The total average power is

$$P_{avg} = F_{rms}^2 R = F_{rms}^2 \text{ since } R = 1\Omega$$

$$P_{av} = F_{rms}^2 = \frac{1}{T} \int_0^T f(t) dt = 0.5A$$

Step 3 of 5

(B) From the Fourier series above

$$|C_0| = \frac{2A}{2}, \quad |C(h)| = \frac{4A}{\left[T_1 (4n^2 - 1) \right]}$$

n	ω_o	$ C(h) $	$2C_n^2$	%POWER
0	0	$\frac{2}{\pi}$	$\frac{4A^2}{(\pi)^2}$	81.7%
1	$2\omega_o$	$\frac{2A}{3\pi}$	$\frac{8A^2}{(9\pi)^2}$	13.01%
2	$4\omega_o$	$\frac{2A}{158\pi}$	$\frac{2A^2}{(225\pi)^2}$	0.72%
3	$6\omega_o$	$\frac{2A}{35\pi}$	$\frac{2A^2}{(1225\pi)^2}$	0.13%
4	$8\omega_o$	$\frac{2A}{63\pi}$	$\frac{3A^2}{(3969\pi)^2}$	0.04%

Step 4 of 5

(C) 31.1%

Step 5 of 5

(D) 0.72%